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Hydraulic valve actuation system with controlled valve seating velocity and method therefor

Abstract

A hydraulic valve actuation system with controlled valve seating velocity and method therefor are proposed. The system includes an actuator with a housing and a drive assembly formed from a drive piston within the housing and an adjustment pin. In an internal combustion engine application, the adjustment pin contacts and moves an engine valve that controls air and fuel entering an engine cylinder. The actuator provides velocity dampening of the drive piston, via an annulus-shaped fluid communication path defined by at least one surface of the drive piston and at least one opposing surface of a damping chamber formed within the housing. The dampening hydraulically cushions the impact of a bottom of the drive piston against a bottom stop of the actuator during opening of the engine valve, which reduces noise and engine wear. In another example, the system is configured to control an engine valve seating velocity.

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit under 35 USC 119(e) of U.S. Provisional Application No. 63/483,281 filed on Feb. 4, 2023, which is incorporated herein by

reference in its entirety.

BACKGROUND OF THE INVENTION

(1) Traditionally, intake and exhaust air valves of a reciprocating internal combustion engine (“ICE”) are operated by a camshaft, which itself is driven by an engine crankshaft. The camshaft typically includes or is formed to provide a separate cam for each engine valve, where the rotation of the camshaft causes each cam to open and close its respective engine valve repeatably over each engine cycle. The engine crankshaft, camshaft and its cams form a valve actuation system.

(2) These existing valve actuation systems have limitations. The systems are rigid, and typically cannot be adjusted to provide optimal timing settings of the valves when engine operating conditions (e.g. speed, load) are changed. This is because the timing of each engine valve opening and closing is set/defined by the cam geometry. Mechanical add-on systems supported by and including electronic controls can be applied to the valve actuation systems to monitor and adjust operation and timing of the valves, but the add-on systems are complicated and expensive.

(3) For full control of the timing of the intake and exhaust valves, hydraulically operated valve actuation systems (“HVA systems”) have been extensively researched in the past. An exemplary HVA system is disclosed in U.S. Pat. No. 12,135,414, issued Jun. 8, 2010. These existing HVA systems are electronically controlled and are designed to replace the camshaft and cams of the existing valve actuation systems. While various HVA systems have been proposed, challenges still remain to make these existing systems commercially viable. The present invention addresses challenges regarding the robustness, durability and quiet operation of the existing HVA systems.

SUMMARY OF THE INVENTION

(4) When an existing HVA system is configured for use in an ICE, in one example, a hydraulic actuator (“actuator”) of the HVA system has an actuation cycle defined by a closed position of the actuator that may correspond to/may be timed with a beginning of the engine cycle, and an open position that may correspond to/may be timed with a middle of the engine cycle.

(5) When used in an ICE, the actuator controls opening and closing of an engine valve. For this purpose, the actuator includes a drive piston that is in contact with an external adjustment pin. The adjustment pin, in turn, connects to the engine valve (or to a bridge, which in turn connects to the engine valve and a second engine valve). Because the actuator provides opening and closing of the engine valve, the open position of the actuator coincides with an open position of the engine valve, and the closed position of the actuator coincides with a closed position of the engine valve. Thus, when the HVA system is used in an ICE, the actuation cycle of the actuator is also known as a valve actuation cycle and may be timed with the engine cycle.

(6) When the ICE is in the middle of its engine cycle (i.e., when the engine valve is fully open/is in its open position to let a mixture of air and fuel to enter an engine cylinder), the actuator is fully open/is in its open position, which also causes the engine valve to be fully open/in its open position. At this time, the drive piston within the actuator is at its lowest position within the actuator, where a bottom surface of the drive piston impacts an inside surface of a bottom stop of the actuator.

(7) A proposed HVA system provides improvements over the existing valve actuation systems, including the existing HVA systems. In one example, each actuator of the proposed HVA system is configured to provide velocity dampening of its drive piston within the actuator when the engine valve is in the middle of its engine cycle. The velocity dampening cushions the impact of the drive piston against the bottom stop, which reduces noise and reduces engine wear. In another example, from the middle of the engine cycle back to a beginning of the cycle, the actuator is configured to control an engine valve seating velocity. In yet another example, the actuator provides velocity dampening of the drive piston within the actuator, as the drive piston nears an inside surface of a top stop of the actuator. When the engine valve is closed/in its closed position, the actuator is in its

closed position, and the engine has returned to the beginning of its (next) engine cycle.

(8) For this purpose, in an embodiment, the drive piston and a housing geometry of the actuator are designed to collectively form a hydraulic damper squeeze pocket and a fluid communication orifice/path in the shape of a ring or annulus as the drive piston approaches at least one of its extreme positions/stops. This annulus-shaped fluid communication path is also known as an annulus. The annulus is located between and defined or otherwise formed by at least one surface of the drive piston, and at least one surface of the actuator housing that opposes the at least one surface of the drive piston. A flow area of the formed annulus is a function of a distance between a surface of the drive piston that defines the flow area, to an inside surface of a hard stop within the actuator that defines the extreme position.

(9) As a result, the proposed HVA system provides low impact velocities for the drive piston within the actuator and for the engine valve on their respective stops, which provides a preferable trajectory of these moving parts as they approach their hard stops.

(10) The proposed HVA system also provides the ability to retrofit existing and new heavy duty industrial ICEs to improve the overall efficiency of the new engines. The proposed system also reduces fuel consumption and carbon output, and is fuel agnostic.

(11) In general, according to one aspect, the invention features a hydraulic actuator. The actuator includes a housing and a drive piston. The housing forms an actuating chamber and a damping chamber within the housing, where the actuating chamber is bounded by the housing and an inside surface of a top stop located at a proximal end of the housing. The damping chamber is bounded by the housing and an inside surface of a bottom stop, where the bottom stop is located at a distal end of the housing that opposes the proximal end.

(12) The drive piston includes a top section and a damping section. The top section is configured to travel within the actuating chamber along a central travel axis that extends from the proximal end to the distal end, and the damping section is configured to travel within the damping chamber along the travel axis. Preferably, at least one surface of the damping section and at least one corresponding surface of the damping chamber are configured to form an annulus-shaped fluid communication path (“annulus”) therebetween that varies in area based upon a location of the damping section relative to the damping chamber, to controllably dampen a velocity of the drive piston during each actuation cycle of the actuator.

(13) In one implementation, the at least one surface of the damping section and the at least one surface of the damping chamber are configured to decrease the area of the annulus as the damping section moves distally towards the inside surface of the bottom stop to dampen the velocity of the drive piston. When a bottom surface of the damping section impacts the inside surface of the bottom stop, in one example, the velocity of the drive piston is less than or equal to two tenths of a meter per second.

(14) The actuator also includes an adjustment pin with a top surface configured to be in unattached contact with a bottom surface of the damping section and to remain so over each actuation cycle of the actuator. For this purpose, in an embodiment, a bottom surface of the adjustment pin that opposes its top surface attaches to and controls opening and closing of an engine valve. In another embodiment, a bottom surface of the adjustment pin interfaces with a bridge, where the bridge controls opening and closing of at least two engine valves.

(15) In another implementation, the at least one surface of the damping section and the at least one surface of the damping chamber are configured to decrease the area of the annulus as the damping section moves proximally toward and nears the inside surface of the top stop, the decrease in area of the annulus decreasing the velocity of the drive piston. Typically, the damping section is substantially disk-shaped, where a shape of the at least one surface of the damping section is a combination of cylindrical and conical shapes. The at least one surface of the damping chamber is also typically cylindrical in shape.

(16) In general, according to another aspect, the invention features a method of operation of a

hydraulic actuator. The actuator includes a housing forming an actuating chamber and a damping chamber within the housing, the actuating chamber being bounded by the housing and an inside surface of a top stop of the actuator, and the damping chamber being bounded by the housing and an inside surface of a bottom stop of the actuator, the top stop being located at a proximal end and the bottom stop being located at a distal end that opposes the proximal end. The method of operation comprises: receiving hydraulic fluid under pressure within the actuating chamber and the damping chamber, the hydraulic fluid driving a drive piston of the actuator, the drive piston including a top section and a damping section, the driving causing the top section to travel in a distal direction within the actuating chamber along a central travel axis that extends from the proximal end to the distal end, and causing the damping section to travel in the distal direction within the damping chamber along the travel axis; and at least one surface of the damping section and at least one corresponding surface of the damping chamber forming an annulus-shaped fluid communication path (“annulus”) therebetween, the annulus varying in area based upon a location of the damping section relative to the damping chamber, the varying in area of the annulus controllably dampening a velocity of the drive piston during each actuation cycle of the actuator.

(17) The method further comprises decreasing the area of the annulus as the damping section moves distally towards the bottom stop, the decrease in area of the annulus dampening the velocity of the drive piston. In one implementation, the method further comprises the velocity of the drive piston being less than or equal to two tenths of a meter per second, upon a bottom surface of the damping section impacting the inside surface of the bottom stop. Preferably, a top surface of an adjustment pin is in unattached contact with a bottom surface of the damping section and remains so over each actuation cycle of the actuator.

(18) In an embodiment, the method further comprises attaching a bottom surface of the adjustment pin that opposes its top surface to an engine valve, the adjustment pin controlling opening and closing of the engine valve. Additionally and/or alternatively, the method further comprises attaching a bridge to a bottom surface of the adjustment pin that opposes its top surface, the bridge controlling opening and closing of at least two engine valves. The method might also further comprise decreasing the area of the annulus as the damping section moves proximally toward and nears the inside surface of the top stop, the decrease in area of the annulus decreasing the velocity of the drive piston.

(19) In general, according to yet another aspect, the invention features a hydraulic valve actuation system (“HVA system”). The HVA system includes an actuator, an adjustment pin and at least one engine valve. The actuator includes a housing that forms a damping chamber within the housing, where the damping chamber is bounded by the housing and an inside surface of a bottom stop located at a distal end of the actuator. The actuator also includes a drive piston including a damping section configured to travel within the damping chamber along a central travel axis.

(20) The damping section is configured to receive hydraulic fluid under pressure. For this purpose, at least one surface of the damping section and at least one surface of the damping chamber are configured to form an annulus-shaped fluid communication path (“annulus”) therebetween. The annulus varies in area based upon a location of the damping section within the damping chamber, to controllably dampen a velocity of the damping section at different locations within the damping chamber during each actuation cycle of the actuator.

(21) The adjustment pin has a top surface configured to be in unattached contact with a bottom surface of the damping section, and the at least one engine valve is controlled by movement of the adjustment pin.

(22) The HVA system might also include a bridge and a second engine valve. Here, a top of the bridge is attached to a bottom surface of the adjustment pin that opposes its top surface, and the engine valve and the second engine valve are attached to a bottom of the bridge that opposes its top. The bridge is preferably configured to control operation of the engine valve and the second engine valve in response to the movement of the adjustment pin.

(23) In the HVA system, in one implementation, the damping section is substantially disk-shaped, and a shape of the at least one surface of the damping section is a combination of cylindrical and conical shapes. In another implementation, the at least one surface of the damping chamber is cylindrical in shape.

(24) The above and other features of the invention including various novel details of construction and combinations of parts, and other advantages, will now be more particularly described with reference to the accompanying drawings and pointed out in the claims. It will be understood that the particular method and device embodying the invention are shown by way of illustration and not as a limitation of the invention. The principles and features of this invention may be employed in various and numerous embodiments without departing from the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the accompanying drawings, reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale; emphasis has instead been placed upon illustrating the principles of the invention. Of the drawings:

(2) FIG. 1 is a schematic diagram that shows the main components and a general layout of an HVA system of an ICE constructed in accordance with principles of the present invention, according to an embodiment, where only one engine valve and one actuator of the HVA system are shown;

(3) FIG. 2 is a cutaway view showing more detail for chambers formed within a housing of the actuator in FIG. 1;

(4) FIG. 3 is a cutaway view of a drive piston of the actuator in the HVA system of FIG. 1;

(5) FIG. 4 is a cutaway view of the actuator in the system of FIG. 1 and shows more detail for the actuator;

(6) FIG. 5A-5C are cutaway views of the actuator that increasingly illustrate its operation shortly after a beginning of an actuator cycle in FIG. 5A, just before a middle of the cycle in FIG. 5B and at the middle in FIG. 5C;

(7) FIG. 6A-6C are cutaway views of the actuator that increasingly illustrate its operation shortly after the middle of the actuator cycle in FIG. 6A, just before the beginning of the cycle in FIG. 6B and at the beginning in FIG. 6C;

(8) FIG. 7 is a cutaway view of an actuator and an engine valve, where an adjustment pin of the actuator is shown in unattached contact with the engine valve;

(9) FIG. 8 is a cutaway view of an actuator, two engine valves and a bridge, where the adjustment pin of the actuator is shown in unattached contact with the bridge, and the bridge is connected to the two engine valves; and

(10) FIG. 9 is a flowchart that describes a method of operation of the actuator.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(11) The invention now will be described more fully hereinafter with reference to the accompanying drawings, in which illustrative embodiments of the invention are shown. This invention may however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

(12) As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the singular forms and the articles “a”, “an” and “the” are intended to include the plural forms as well, unless expressly stated otherwise. It will be further understood that the terms: includes, comprises, including and/or comprising, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components,

but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Further, it will be understood that when an element, including component or subsystem, is referred to and/or shown as being connected or coupled to another element, it can be directly connected or coupled to the other element or intervening elements may be present.

(13) It will be understood that although terms such as “first” and “second” are used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. Thus, an element discussed below could be termed a second element, and similarly, a second element may be termed a first element without departing from the teachings of the present invention.

(14) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(15) FIG. 1 shows components and a general layout of an HVA system **100**, according to an embodiment. In the illustrated example, the HVA system **100** is configured for operation with an ICE **70** and includes one or more hydraulic control valves **140**, an actuator **20**, a pressure regulator and relief valve **252**, a hydraulic pump **120**, a reservoir **250**, a hydraulic accumulator **200**, an adjustment pin **24** and an engine valve **30**.

(16) In the illustrated example, only a portion of the ICE **70** in connection with the engine valve **30** is shown. The actuator **20**, ICE **70**, adjustment pin **24** and engine valve **30** are shown in cross section without cross-hatching to show more detail for components therein. While a separate actuator **20** is typically employed to control operation of each engine valve **30**, only one actuator **20**/engine valve **30** pair is shown. In a similar vein, while a separate hydraulic control valve **140** is typically employed to control operation of each actuator **20**, only one control valve **140** is shown.

(17) More detail for the components of the HVA system **100** is as follows. The reservoir **250** stores fluid (e.g., hydraulic or engine oil) under low pressure, typically at 0 pounds per square gauge (psig). The hydraulic pump **120** has an input side **93** and an output side **95**, and the pressure regulator and relief valve **252** has an input side **96** and an output side **97**.

(18) The ICE **70** includes a cylinder head **78** and an engine cylinder **74** located below the cylinder head **78**. The engine cylinder **74** includes an engine piston **72** that travels within the engine cylinder **74**. A valve seat **38** is located between the cylinder head **78** and the engine cylinder **74** and provides an opening into the engine cylinder **74**. At a top **109** of the cylinder head **78**, a valve pocket **79** is formed within the cylinder head and has a pocket floor **17**.

(19) The actuator **20** includes various components. These components include a housing **26** that forms a damping chamber **50** and an actuating chamber (not visible) within the housing **26**, a top stop **42**, a bottom stop **52**, a drive piston **10** and a boost piston **22**. The top stop **42** has an inside surface **27** and is located at a proximal end **21** of the actuator **20**. The bottom stop **52** has an inside surface **87** and is located at a distal end **31** of the actuator **20** that opposes the proximal end **21**. The actuator **20** also includes a vent port **44** that connects to the damping chamber **50** and extends through a side of the housing **26**, and includes a bidirectional fluid port **57** incorporated within the top stop **42**. The drive piston **10** includes a top section **32** and a damping section **34**.

(20) The engine valve **30** includes a stem **33** and a head **35** connected to the stem **33**. A collar **37** affixed to the valve stem **33** and a valve spring **36** are also shown.

(21) The housing **26** of the actuator **20**, the adjustment pin **24**, the boost piston **22** and the drive piston **10** are typically each formed from a unitary piece of metal or hardened metal alloy material such as stainless steel. However, aluminum alloys can also be used. The actuator **20** is substantially cylindrical in shape.

(22) The drive piston **10** and the boost piston **22** are actuating pistons that operate together to move the engine valve **30** downward/in a distal direction, via the adjustment pin **24**. The boost piston **22** has a much shorter stroke than the drive piston **10**, and the boost piston **22** accelerates the entry of the engine valve **30** into the engine cylinder **74** with a relatively low amount of additional high pressure hydraulic fluid.

(23) The HVA system **100** is generally arranged as follows. With respect to hydraulic-related connections in the HVA system, the reservoir **250** stores fluid (e.g., hydraulic or engine oil) and connects to the input side **93** of the hydraulic pump **120** and the output side **97** of the pressure regulator and relief valve **252**. The output side **95** of the pump **120** connects to and provides hydraulic fluid under pressure to the hydraulic control valve **140**. This connection is also known as a pressurized path **13**.

(24) The hydraulic control valve **140** is an electromechanical switch that has separate connections to the input side **96** of the pressure regulator and relief valve **252**, to the pump **120** and to the fluid port **57** of the actuator **20**. The connection to the pressure regulator and relief valve **252** is along the low pressure return path **15**, while the connection to the pump **120** is along the pressurized path **13**. The hydraulic control valve **140** is under computer control and switchedly connects the pump **120** or the pressure regulator and relief valve **252** to the fluid port **57**.

(25) The hydraulic accumulator **200** connects to the low pressure return path **15** and provides pressure to the pressure regulator and relief valve **252** via its input side **96**, and the vent port **44** connects to and vents to the low pressure return path **15**.

(26) A top surface **82** of the adjustment pin **24** is designed to be in unattached contact with the drive piston **10** throughout each engine cycle/actuator cycle. In more detail, a top surface **82** of the adjustment pin **24** is in contact with the bottom surface **54** of the damping section **34**. A bottom surface **84** of the adjustment pin **24** that opposes its top surface **82** is attached to a top of the stem **33** of the engine valve **30**. The valve spring **36** seats within the valve pocket **79** and rests upon the pocket floor **17**, and is disposed between the collar **37** and the pocket floor **17**.

(27) A central travel axis **99** passes through the actuator **20** from its proximal end **21** to its proximal end **31**. The travel axis **99** also passes centrally through the fluid port **57**, the drive piston **10**, the actuating chamber **60**, the damping chamber **50**, the adjustment pin **24** and the engine valve **30**.

(28) The head **35** of the engine valve **30** seats within the valve seat **38** when the engine is at the beginning of its engine cycle (as shown). The valve seat **38** has an opening (not shown, it is covered by the head **35** in the figure) that opens into the engine cylinder **74**.

(29) The HVA system **100** generally operates as follows. Here, the actuator cycle of the actuators **20** is timed to correspond to the engine cycle of the ICE **70**. Each hydraulic control valve **140** is electronically controlled and uses hydraulic fluid, which is stored in the reservoir **250** typically at or slightly above ambient pressure. In one example, the hydraulic fluid is separate from engine lube oil of the ICE **70**. In another example, the engine lube oil also functions as the hydraulic fluid.

(30) At its input side **93**, the pump **120** draws hydraulic fluid from the reservoir **250** and pressurizes the fluid in a range typically between 1000 and 4000 psig. A pressurized flow of hydraulic fluid from the pump **120** to the hydraulic control valve **140** and to the actuator **20**, over the pressurized path **13**, is indicated by arrows with reference number **11**.

(31) Each actuator **20** is in a normally closed position, where its drive piston **10** is located at a topmost position within the housing **26**. The hydraulic control valve **140**, when activated/turned on, connects to the pressurized path **13**. This enables the pressurized flow **11** of hydraulic fluid to enter the fluid port **57** of the actuator **20**. This pressurized fluid exerts a downward force/a force in a distal direction upon the drive piston **10**, which in turn causes the drive piston **10** to move downward within the chambers of the housing **26**. The adjustment pin **24** moves downward in response, the result of which moves the engine valve **30** downward. The adjustment pin **24** is designed to accurately position the drive piston **10** when the engine valve **30** is seated within the valve seat **38** of the engine cylinder **74**.

(32) When the engine reaches a middle of its cycle, the damping section **34** of the drive piston **10** is located at its most extreme downward position within the damping chamber **50**, and the engine valve **30** is in an open position within the engine cylinder **74**.

(33) When the hydraulic control valve **140** is deactivated/turned off, the engine **70** has just begun to move from the middle of its cycle back to the beginning of its engine cycle, and the drive piston **10** is no longer under pressure. As a result, hydraulic fluid can flow back from the actuator **20** out the fluid port **57** and through the hydraulic control valve **140** into the reservoir **250**, along the return path **15**. This return flow of the hydraulic fluid is indicated by an arrow with reference **12**. A relatively low pressure of about 40-60 psig is maintained in the return path **15** via the pressure regulator and relief valve **252** and the hydraulic accumulator **200** to help reduce pressure spikes in the HVA system **100**.

(34) Also when the hydraulic control valve **140** is turned off, because the actuator **20** is no longer under pressure, the actuator **20** requires an external force to move the drive piston **10** and the engine valve **30** back to their original/closed positions. In an ICE **70**, the force that returns the drive piston **10** and the engine valve **30** back to their original/closed positions, and helps pump spent hydraulic actuation fluid from the actuator **20** into the return path **15**, is provided by the valve spring **36**. Any leakages from the actuator **20** are collected and returned back into the return path **15** via the hydraulic accumulator **200**.

(35) As a result, the on/off switch timing of the hydraulic control valves **140** allows the opening and closing time of the intake/exhaust valves **30** to be optimal. This timing is provided by an electronic control unit (ECU) of the HVA system **100** (not shown), which connects to each of the hydraulic control valves **140**. The timing is specified by a computer program (e.g., software or firmware) that executes on a processor of the ECU.

(36) It can also be appreciated that dynamic seals, check valves, or secondary solenoid valves are not required in the HVA system **100**. Because the HVA system **100** has fewer moving parts than that of existing valve actuation systems, experimentation has shown improved cycle to cycle consistency across engine cycles of ICEs **70** that incorporate the HVA system **100**, as compared to ICEs with existing valve actuation systems.

(37) FIG. 2 shows more detail for an actuating chamber **60** and the damping chamber **50** formed within the housing **26** of the actuator **20**. In the figure, the actuator **20** is shown in cross-section with its drive piston **10** and boost piston **22** removed.

(38) The actuating chamber **60** is centrally aligned within the housing **26**, along the travel axis **99**. The actuating chamber **60** is bounded by the housing **26** and the inside surface **27** of the top stop **42**. A bore **61** within the top stop **42** allows the input port **57** to seat within the top stop **42** and to extend into the actuating chamber **60**. The actuating chamber **60** is located above and extends distally into/connects to the damping chamber **50**.

(39) The damping chamber **50** has a top inside surface **66** and is also centrally aligned within the housing **26**, along the travel axis **99**. The damping chamber **50** is bounded by the housing **26** and the inside surface **87** of the bottom stop **52**. A bore **59** within the bottom stop **52** allows the adjustment pin **24** (not shown) to enter/extend into the damping chamber **50**, and to exit from the damping chamber **50**. The adjustment pin **24** is designed to tightly fit into the bore **59** to minimize fluid leakages from the damping chamber **50**.

(40) The damping chamber **50** has an upper chamber surface **101** and a lower chamber surface **104** formed within. These surfaces **101**, **104** are generally cylindrical in shape. The upper chamber surface **101** is located below and adjacent to the top inside surface **66** of the damping chamber **50**. The lower chamber surface **104** is located above and adjacent to the inside surface **87** of the bottom stop **52**. While only a portion of the upper and lower chamber surfaces **101**, **104** are shown, these surfaces are provided around the entirety of the damping chamber **50**.

(41) During operation of the actuator **20**, the upper chamber surface **101** is designed to oppose an upper damping surface of the drive piston **10**, and the lower chamber surface **104** is designed to

oppose a lower damping surface of the drive piston **10**. The upper chamber surface **101** and its corresponding upper damping surface of the drive piston **10**, and the lower chamber surface **104** and its corresponding lower damping surface of the drive piston **10**, enable controlled deceleration of the drive piston **10** as a function of the location of the drive piston **10** relative to the damping chamber **50**. More details regarding these surfaces are provided in the descriptions of FIGS. **3**, **4**, **5A-5C** and **6A-6C**, included hereinbelow.

(42) FIG. **3** is a cutaway view of the drive piston **10**, shown in vertical cross-section, according to an embodiment. Here, the cross-section of the drive piston **10** resembles an upside-down letter “T”.

(43) The top section **32** is substantially cylindrical in shape, while the damping section **34** is wider than the top section and is substantially disk-shaped. The damping section **34** has a much larger diameter than the top section **32**, and is designed to dampen motion of the actuator **20** and engine valve **30** as they approach their extreme positions during each engine cycle.

(44) Because the damping section **34** is substantially disk-shaped, when viewed in cross-section as shown, it is substantially octagonal in shape with eight sides **s1** through **s8**. A top side **s1** and a bottom side **s2** oppose one another, are substantially horizontal and parallel to one another and have substantially the same length. Opposing sides **s4** and **s7** of the damping section **34** are substantially vertical and parallel to one another and have substantially the same length. Side **s3** tapers inward towards top side **s1**, and side **s6** tapers inward toward top side **s1**. In a similar vein, side **s5** tapers inward towards bottom side **s2**, and side **s8** tapers inward toward bottom side **s2**. The sides **s3**, **s4** and **s5** (or the sides **s6**, **s7** and **s8**) collectively form a “shoulder” **90** of the damping section **34**.

(45) An upper damping surface **102** and a lower damping surface **103** of the damping section **34** of the drive piston **10** are shown. The upper damping surface **102** is located below and is adjacent to a top **56** of the damping section **34**. The lower damping surface **103** is located above and is adjacent to the bottom surface **54** of the damping section **34**. While only a portion of the upper and lower damping surfaces **102**, **103** are shown, these surfaces are provided around the entirety of the damping section **34**.

(46) In one example, the upper and lower damping surfaces **102**, **103** are a combination of cylindrical and conical surfaces. In other examples, the upper and lower damping surfaces **102**, **103** might be radii or curves. As noted in the description of FIG. **2**, the upper damping surface **102** is designed to oppose the upper chamber surface **101** of the damping chamber **50**, while the lower damping surface **103** is designed to oppose the lower chamber surface **104** of the damping chamber **50**.

(47) FIG. **4** shows more detail for the actuator **20**, shown in cross-section. In the illustrated example, the actuator **20** is in a closed position. When the actuator cycle of the actuator is timed to an engine cycle of an ICE **70**, the actuator is in its closed position when the engine **70** is at the beginning of each engine cycle.

(48) Also at the beginning of the engine cycle, the head **35** of the engine valve **30** is in its seated position/is seated within the valve seat **38** of the engine cylinder **74**. Because the drive piston **10** and the adjustment pin **24** are configured to move within the actuating chamber **60** and the damping chamber **50** to drive the engine valve **30**, the combination of the drive piston **10** and the adjustment pin **24** are also known as a drive assembly.

(49) Only a portion of the stem **33** of the engine valve **30** is shown. The adjustment pin **24** is designed to be in mating contact with both the drive piston **10** and the engine valve **30**. The adjustment pin **24** is arranged vertically and is substantially centered with respect to the travel axis **99**.

(50) The boost piston **22** and the top section **32** of the drive piston **10** are seated within the actuating chamber **60**. The damping section **34** is seated and designed to travel within the damping chamber **50**. The top section **32** is guided through the actuating chamber **60** along the travel axis **99** during operation of the actuator **20**.

(51) The surfaces **101**, **104** of the damping chamber **50** and the surfaces **102**, **103** of the damping

section **34** of the drive piston **10** are arranged relative to one another as follows. At the beginning of the engine cycle, the actuator **20** is in its closed position (as shown). The drive piston **10** is at its top-most position within the actuator **20**, and the upper chamber surface **101** of the damping chamber **50** and its corresponding upper damping surface **102** of the damping section **34** directly oppose one another.

(52) Also when the actuator **20** is in its closed position, a slight gap **98** is maintained between the top inside surface **66** of the damping chamber **50** and the top **56** of the damping section **34**. The gap **98** is needed to account for thermal expansion of the valve stem **33**. Due to a tolerance stack-up, this gap **98** will not be accurately controlled without adjusting the length of the adjustment pin **24**.

(53) In a middle of the engine cycle, the drive piston **10** is at its bottom-most position within the actuator **10**, and the actuator **20** is fully open. The bottom surface **54** of the damping section **34** is located just above the inside surface **87** of the bottom stop **52**, and the lower chamber surface **104** of the damping chamber **50** and its corresponding lower damping surface **103** of the damping section **34** directly oppose one another.

(54) An important aspect of engine valve operation that affects the reliability/durability and noise of the HVA system **100** is how fast the actuators **20** and engine valves **30** are moving as they approach their fully open and fully closed positions. The fully open position of each engine valve **30** is preferably met by a hard stop, meaning the drive piston **10** or the engine valve **30** contacts a mechanical stop where metal-to-metal contact ensures an accurately controlled open position for the engine valve **30**. Without such a hard stop, the open position of the engine valve **30** is difficult to control, and cylinder-to-cylinder variability in this open position leads to detrimental cylinder-to-cylinder variability of engine valve closing timing.

(55) In the fully closed position of each engine valve **30**, the metal-to-metal contact to limit the motion of the engine valve **30** is even more important because this contact is required for the engine valve **30** to fully seal/seat within the valve seat **38**. If the drive piston **10** or the engine valve **30** establish contact with their respective stops at too high a velocity, unacceptable wear of the contact surfaces and/or unacceptably high noise may occur. The HVA system **100**, via its actuator **20**, controllably reduces the velocity of the drive piston **10** and engine valve **30** prior to impact with their stops.

(56) Operation of the actuator **20** over a full engine cycle/valve actuation cycle (opening and closing) is generally as follows. At the beginning of the cycle, the drive piston **10**, the adjustment pin **24** and the stem **33** of the engine valve **30** are in their top-most positions relative to the actuator **20**. The head **35** of the engine valve **30** is also seated within the valve seat **38** such that pressure is not able to enter into the engine cylinder **70** within which the engine piston **72** travels.

(57) When the control valve **140** is turned on, hydraulic fluid under pressure enters the fluid port **57**. As a result, the top of the boost piston **22** and the drive piston **10** are placed under hydraulic pressure. The boost piston **22** and the drive piston **10**, together with the adjustment pin **24**, begin to move downward/distally towards the distal end **31** of the actuator **20** along the travel axis **99**. The damping section **34** of the drive piston **10** begins to push the adjustment pin **24** downward, which in turn begins to push the engine valve **30** into an open position. When in its open position, the engine valve **10** enables a mixture of compressed air and fuel to enter the engine cylinder **74** via the valve seat **38**.

(58) When the engine reaches the middle of its cycle, the bottom surface **54** of the damping section **34** impacts a top of the bottom stop **52**, the engine valve **30** is in its fully open position, and the valve spring **36** is at maximum compression. At this point in the engine cycle, the control valve **140** is turned off, and hydraulic fluid under pressure no longer enters the fluid port **57** until the beginning of the next engine cycle.

(59) Because the drive piston **10** is not under pressure from the middle of the engine cycle/actuation cycle back to the beginning of the next engine cycle, the force provided by the compressed spring **36** provides an upward force upon the collar **37**. In response, the engine valve

30 begins to move upward/move proximally toward the top stop **42**. Consequently, the adjustment pin **24** begins to move upward, which in turn causes the drive piston **10** to move upward. This upward/proximal motion continues until the top **56** of the damping section **34** reaches the gap **98**. At this point, the actuator **20** and the engine valve **30** have returned to their fully closed positions, and the top surface **56** of the damping section **34** and the top inside surface **66** of the damping chamber **50** do not come into contact with one another/are separated by the gap **98**. After this point, the control valve **140** is turned on, which starts the next engine cycle.

(60) FIG. 5A-5C illustrate operation of the actuator **20** shortly after the beginning of its actuator cycle and until the middle of the actuator cycle. In the description below, the contributions of the chamber surfaces **101**, **104** and the damping surfaces **102**, **103** during operation are emphasized.

(61) While the actuator cycle is often tied to/coincides with the engine cycle, it can also be appreciated that the ECU can instruct the hydraulic control valves **140** to open and close their respective actuators **20** independently of the engine cycle. In examples, some engine cylinders **74** are routinely not fired to save energy, and the actuators **20** can be opened and closed to correspondingly open the intake and exhaust engine valves **30** at times more suitable for exhaust gas recirculation (EGR) purposes, or for other situations.

(62) In FIG. 5A, the lower damping surface **103** starts to get squeezed into a 'pocket', or "squeeze pocket" **107** that is defined by a bottom inner diameter "d" of the housing **26** and the top surface **87** of the bottom stop **52**. The volume of the squeeze pocket **107** decreases as the drive piston **10** further travels downward/distally in the direction indicated by arrow A, along the travel axis **99** towards the bottom stop **52**. Since the entire inner volume of the actuator **20** is filled with hydraulic fluid, this nearly incompressible fluid has to be squeezed out through the annulus formed between the lower damping surface **103** and the lower chamber surface **104**.

(63) Arrows B indicate the flow of hydraulic fluid through the annulus. Because the area of the annulus is quite small, the hydraulic fluid can only escape at a sufficient rate if the pressure rises in the squeeze pocket **107**. As the area of the annulus decreases, the resultant rising pressure in the squeeze pocket **107** exerts a significant upward/proximally directed force upon the bottom surface **54** of the damping section **34**, which will slow down the downward motion of the drive piston **10**/decrease its velocity.

(64) In FIG. 5B, the drive piston **10** is still moving downward, but at this position the conical part of the damping surface **103** is already located within the squeeze pocket **107**. As a result, the cylindrical part of the damping surface **103** starts to get squeezed into the squeeze pocket **107**. The annulus is now tighter/its area is at its smallest value, and the flow area will not change any further.

(65) In FIG. 5C, the drive piston **10** has reached its full travel length, corresponding to a middle of the engine cycle/actuation cycle of the actuator **20**. Here, the bottom surface **54** of the damping section **34** is now in contact with the inside surface **87** of the bottom stop **52**. Due to the formation of the squeeze pocket **107**, the drive piston **10** slows to a very low velocity (typically less than or equal to two tenths of a meter per second) before its bottom surface **54** establishes contact with the inside surface **87** of the bottom stop **52**. This is also known as an impact velocity of the drive piston **10**/actuator **20**. In another example, the impact velocity is less than or equal to 1/10 meters per second. In still another example, the impact velocity is less than or equal to 1 meter per second. The minimized impact velocity provided by the actuator **20** results in a quiet and non-destructive impact of the bottom surface **54** of the damping section **34** upon the inside surface **87** of the bottom stop **52**. Such a feature not only reduces engine noise as compared to traditional engines, but also minimizes wear upon the actuator **20**, thus extending engine life/improving reliability.

(66) In addition to the minimized impact velocity, the rate at which the drive piston **10** slows down prior to impact is also of importance. One design consideration is that the deceleration should be as rapid as possible so that the engine valve **30** reaches high lift as soon as possible, for better breathing of the engine. At the same time, if the deceleration is too rapid/occurs over too short an interval, the associated momentum will carry on the adjustment pin **24** and the engine valve **30**, and

the adjustment pin **24** may separate from the drive piston **10** (see FIG. 5B).

(67) Another drawback of drive piston **10** deceleration occurring over too short a time interval is that the adjustment pin **24** will be pushed back with an increased and undesirable force against the drive piston **10** by the valve spring **36**. This impact can also generate high noise and component wear. Therefore, any separation between the adjustment pin **24** and the drive piston **10** should altogether be avoided. By carefully selecting the height of the lower chamber surface **104**, the height of the conical section on the lower damping surface **103**, and the diameters defining the geometry of the resulting annulus, the deceleration can be set to be as fast as possible, while also avoiding separation between the adjustment pin and the drive piston during operation of the actuator, and also providing low seating velocity of the drive piston on the bottom stop.

(68) FIG. 6A-6C similarly illustrate the damping mechanism provided by the actuator **20** when the engine valve **30** goes through its closing process/when the engine transitions from the middle of its engine cycle back to its beginning. The engine valve **30** should impact its valve seat **38** at a low velocity, again for noise and wear reduction. Separation between the moving parts this time is not an issue because the part on which the decelerating force is applied (the drive piston **10**) is the component that decelerates the rest of the drive assembly. At the same time, the rate at which the engine valve **30** approaches its valve seat **38** can be important. For example, and for reasons not detailed herein, it can be advantageous to keep the engine valve slightly open for a few milliseconds before complete closure.

(69) In FIG. 6A, the actuator **20** is configured for closing of the engine valve **30** in a substantially similar way as that used for the opening of the engine valve in FIG. 5A. Here, the drive assembly moves upward in a direction indicated by arrow C, along the travel axis **99**. A squeeze pocket **107** is formed when the drive piston **10** reaches the position as shown in the figure. Here, the upper chamber surface **101** and its corresponding upper damping surface **102** now define the annulus through which the hydraulic fluid will be squeezed out, the direction of which is indicated by arrows D. This generates relatively high pressure in the squeeze pocket **107**. The annulus flow area varies with the drive piston **10** position according to the actual shape of the surfaces **101** and **102**.

(70) In the illustrated example, in a manner similar to how the lower chamber surface **104** and the lower damping surface **103** are employed during open side damping, the upper damping surface **102** is a cylindrical surface and the upper chamber surface **101** is a combination of a conical and a cylindrical surface. With this setup, the annulus flow area is first larger and later it is smaller.

(71) In FIG. 6B, the drive piston **10** is still moving upward, now at an already reduced velocity, and it is partially squeezed into the squeeze pocket **107** to the point that the constant diameter cylindrical part of the upper damping surface **102** begins to engage with the squeeze pocket **107**. The annulus area remains substantially the same as the drive piston **10** moves further upward towards the proximal end **21** of the actuator **20**.

(72) FIG. 6C illustrates the position of the actuator **20** and drive assembly (i.e., engine valve **30** is now closed/seated) when the engine cycle is complete and has transitioned back to the beginning of the cycle. As described previously, the height of the squeeze pocket **107**, the height of the conical upper chamber surface **101** and the diameters involved can be selected such that optimal closing trajectory of the engine valve **30** is achieved.

(73) More details for the HVA system **100** can be appreciated. The annulus is formed by a female part (here, the chamber surfaces **101** and **104**) having a cylindrical surface, and a male part (here, the damping surfaces **102** and **103**) having a combination of a cylindrical and a conical surface. Also, the height of the conical surface is less than the height of the pocket into which the conical surface is squeezed. This is true for both the damping provided during opening of the actuator **20** and the damping provided during closing of the actuator **20**. However, the surfaces **101/102** and **103/104** that define the annulus can be manufactured to other dimensions and other shapes as well.

(74) The optimum choice for the shape and dimensions of these surfaces **101/102** and **103/104** depends on a number of factors such as the size of the HVA system **100** and its actuators **20**, the

engine operating speedrange, a typical cylinder pressure range, and the like. Most importantly, these surfaces **101/102** and **103/104** are configured to provide an annulus in which the flow area varies as the drive piston **10** and engine valve **30** approach a hard stop to provide an optimum slow-down trajectory for the parts of the drive assembly.

(75) The HVA system **100** might also be configured with or without use of the boost piston **22**. The use of the boost piston **22** is typically advantageous, but it is not required.

(76) FIG. 7 shows an actuator **20**, an adjustment pin **24** in contact with the actuator **20** and an engine valve **30**. The adjustment pin **24** is in unattached contact with the drive piston **10** of the actuator **20**. Specifically, the top surface **82** of the adjustment pin **24** contacts the bottom surface **54** of the damping section **34**. The bottom surface **84** of the adjustment pin **24**, in turn, connects to the top of the stem **33** of the engine valve **30**. The collar **37** of the valve stem **33** is not shown in the figure.

(77) FIG. 8 shows an actuator **20**, an adjustment pin **24** in contact with the actuator **20** and an engine valve **30** as in FIG. 7. However, a bridge **802** is inserted between the bottom surface **84** of the adjustment pin **24** and the top of the stem **33**. The bridge **802** is shaped somewhat similar to an upside-down letter “U” and has a flat top surface **805** and two flat bottom surfaces **803-1** and **803-2**. In one embodiment, as shown, the two flat bottom surfaces **803-1** and **803-2** are recessed within a body of the bridge **802**.

(78) In more detail, the top surface **805** of the bridge **802** is attached to the bottom surface **84** of the adjustment pin **24**. The bottom surfaces **803-1**, and **803-2** of the bridge **802**, in turn, connect to two engine valves **30-1** and **30-2**, respectfully. In this way, when the actuator **20** is in its open position (such as when an ICE **70** is in the middle of its engine cycle), the actuator **20**, via the bridge **802**, can push/move the two engine valves **30-1**, **30-2** simultaneously.

(79) The actuator **20** is designed so that the top surface **82** of the adjustment pin **24** is and remains in unattached contact with the bottom surface **54** of the damping section **34** at all times during each actuation cycle of the actuator **20**. This is a result of the velocity damping of the drive position **10**, enabled by the chamber surfaces **101**, **104** and the damping surfaces **102**, **103** and the annulus that the surfaces provide or otherwise form during operation of the actuator **20**. The gradual reduction in the velocity of the drive piston **10** assures that the drive piston **10**, the adjustment pin **24**, engine valve **30**, and optionally the bridge **802** and its engine valves **30-1**, **30-2** do not mechanically separate in a spring return-based HVA system **100** as in FIG. 1.

(80) FIG. 9 is a flowchart that describes a method of operation of the HVA system **100**. The method starts at step **902**.

(81) In step **902**, the actuator **20** receives hydraulic fluid under pressure that fills the actuating chamber **60** and the damping chamber **50**. According to step **904**, the hydraulic fluid provides fluid pressure upon the drive piston **10** to drive the drive piston **10** within the actuating chamber **60** and the damping chamber **50** along the central axis **99**, the latter of which extends from the proximal end **21** to the distal end **31** of the actuator **20**. The drive piston **10** includes a top section **32** and a damping section **34**. The drive piston **10** travels in a distal direction in response to receiving the hydraulic fluid under pressure. The top section **32** travels within the actuating chamber **60** and the damping section **34** travels within the damping chamber **50** along the central axis **99**.

(82) In step **906**, at least one surface **102**, **103** of the damping section **34** and at least one opposing surface **101**, **104** of the damping chamber **50** form an annulus-shaped fluid communication path (“annulus”) between the damping section **34** and the damping chamber **50**, where the annulus varies in area based upon a location of the damping section **34** relative to the damping chamber **50**. Here, the varying area of the annulus controllably dampens a velocity of the drive piston **10** within the damping chamber **50** during each actuation cycle of the actuator **20**.

(83) In step **908**, the area of the annulus decreases as the damping section **34** moves distally toward the inside surface **87** of the bottom stop **52** to effect a closed position of the actuator **20**, where the decrease in area of the annulus decreases the velocity of the drive piston **10**. In this example, the

lower chamber surface **104** and its corresponding lower damping surface **103** of the drive piston **10** typically define the annulus.

(84) Then, in step **910**, upon the bottom surface **54** of the damping section **34** impacting the inside surface **87** of the bottom stop **52**, in one implementation, the velocity of the drive piston **10** is equal to or less than two tenths of a meter per second. In this example, the lower chamber surface **104** and its corresponding lower damping surface **103** of the drive piston **10** typically define the annulus.

(85) Finally, in step **912**, the area of the annulus decreases as the damping section **34** moves proximally toward and nears the inside surface **27** of the top stop **42** to effect an open position of the actuator **20**, where the decrease in area of the annulus decreases the velocity of the drive piston **10**. In this example, the upper chamber surface **101** and its corresponding upper damping surface **102** of the drive piston **10** typically define the annulus.

(86) The fuel and hydraulic fluid pressures specified in this disclosure are typical, but they can be higher or lower in a specific implementation of the HVA system **100**. It can also be appreciated that the principles of the HVA system **100** can be applied to any hydraulic actuator application when a need for the drive piston **10** of the actuator **20** to approach its extreme positions at low and controlled velocity is desirable.

(87) While the proposed HVA system **100** is preferably tailored to provide control of engine valves in ICEs, it can also be appreciated that the HVA system and its components have other industrial applications. In examples, the actuator **20** might be connected to a hydraulic system and configured to control robotic arms and conveyor systems in a manufacturing facility, and one or more components of machine presses and computer numerical control (CNC) machining systems. In other examples, the actuator **20** might be deployed within a vehicle such as an airplane or automobile and connected to its hydraulic system. The actuator **20** might also be configured to control operation of ailerons and flaps in airplanes, and components in braking and regenerative systems in vehicles such as ICE-only vehicles, hybrid ICE/electric and electric-only vehicles, in examples.

(88) It can also be appreciated that instead of using the valve spring **36** to return the engine valve **30** and the actuator drive pistons **10** to their initial positions, hydraulic pistons might also be used for this purpose. In this way, the hydraulic pistons enable the actuator **20** to be used in applications other than ICEs **70**.

(89) While this invention has been particularly shown and described with references to preferred embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

Claims

1. A hydraulic actuator, comprising: a cylindrical housing including: a top stop, a bottom stop, an actuating chamber defined by the housing and an inside surface of the top stop so as to extend along a central axis of the housing, and a damping chamber defined by the housing and an inside surface of the bottom stop so as to extend along the central axis; and a hydraulically-actuated drive piston including: a top section formed at a first end of the drive piston, the top section configured to reciprocate within the actuating chamber along the central axis, and a damping section formed at a second end of the drive piston opposite the first end, the damping section configured to reciprocate within the damping chamber along the central axis; wherein an annulus consisting of an annular fluid communication path is formed between at least one surface of the damping section and at least one surface of the damping chamber such that an area of the annulus varies based on a location of the damping section relative to the damping chamber so as to controllably dampen a velocity of the drive piston during each actuation cycle of the actuator, wherein a diameter of the

damping section is greater than a diameter of the top section, wherein a diameter of the damping chamber is greater than a diameter of the actuating chamber so as to accommodate the damping section, and wherein the drive piston is included entirely within the housing such that, during operation of the drive piston, the damping section remains in the damping chamber, and the top section is at least partially disposed in the actuating chamber.

2. The actuator of claim 1, wherein the annulus is formed between at least one lower surface of the damping section and at least one lower surface of the damping chamber such that the area of the annulus decreases as the damping section approaches the inside surface of the bottom stop so as to dampen the velocity of the drive piston.
3. The actuator of claim 1, wherein the velocity of the drive piston is less than or equal to 0.2 m/s when the damping section impacts the inside surface of the bottom stop.
4. The actuator of claim 1, further comprising an adjustment pin including a top surface configured to remain in unattached contact with a bottom surface of the damping section throughout each actuation cycle of the actuator.
5. The actuator of claim 4, wherein the adjustment pin further includes a bottom surface configured to be attached to an engine valve so as to control opening and closing of the engine valve.
6. The actuator of claim 4, wherein a bottom surface of the adjustment pin engages a bridge configured to control opening and closing of at least two engine valves.
7. The actuator of claim 1, wherein the annulus is formed between at least one upper surface of the damping section and at least one upper surface of the damping chamber such that the area of the annulus decreases as the damping section approaches a top inside surface of the damping chamber so as to dampen the velocity of the drive piston.
8. The actuator of claim 1, wherein the damping section is disk-shaped, and wherein the at least one surface of the damping section is a conical surface.
9. The actuator of claim 1, wherein the at least one surface of the damping chamber is a cylindrical surface.
10. A method of operating the hydraulic actuator of claim 1, the method comprising: receiving a flow of pressurized hydraulic fluid within the actuating chamber via the top stop so as to drive the top section of the driving piston towards the damping chamber; biasing the damping section towards a top inside surface of the damping chamber when the flow of pressurized hydraulic fluid is stopped; varying the area of the annulus based on the location of the damping section relative to the damping chamber so as to controllably dampen the velocity of the drive piston during each actuation cycle of the actuator; and dampening the velocity of the drive piston via the annulus as the damping section alternately approaches the inside surface of the bottom stop and the top inside surface of the damping chamber.
11. The method of claim 10, wherein the area of the annulus decreases as the damping section approaches the inside surface of the bottom stop so as to dampen the velocity of the drive piston.
12. The method of claim 10, wherein the velocity of the drive piston is less than or equal to 0.2 m/s when the damping section impacts the inside surface of the bottom stop.
13. The method of claim 10, wherein the hydraulic actuator further comprises an adjustment pin including a top surface configured to remain in unattached contact with a bottom surface of the damping section throughout each actuation cycle of the actuator.
14. The method of claim 13, wherein the adjustment pin further includes a bottom surface configured to be attached to an engine valve so as to control opening and closing of the engine valve.
15. The method of claim 13, wherein a bottom surface of the adjustment pin engages a bridge configured to control opening and closing of at least two engine valves.
16. The method of claim 10, wherein the area of the annulus decreases as the damping section approaches the top inside surface of the damping chamber so as to dampen the velocity of the drive piston.

17. A hydraulic valve actuation (HVA) system, comprising: a hydraulic actuator including: a cylindrical housing including: a top stop, a bottom stop, an actuating chamber defined by the housing and an inside surface of the top stop so as to extend along a central axis of the housing, and a damping chamber defined by the housing and an inside surface of the bottom stop so as to extend along the central axis; a hydraulically-actuated drive piston including: a top section formed at a first end of the drive piston, the top section configured to reciprocate within the actuating chamber along the central axis, and a damping section formed at a second end of the drive piston opposite the first end, the damping section configured to reciprocate within the damping chamber along the central axis; an adjustment pin including a top surface configured to remain in unattached contact with a bottom surface of the damping section throughout each actuation cycle of the actuator; and at least one engine valve configured to be driven via movement of the adjustment pin; wherein the damping chamber is configured to receive pressurized hydraulic fluid, wherein an annulus consisting of an annular fluid communication path is formed between at least one surface of the damping section and at least one surface of the damping chamber such that an area of the annulus varies based on a location of the damping section relative to the damping chamber, wherein the annulus is configured to dampen a velocity of the drive piston as the damping section alternately approaches the inside surface of the bottom stop and a top inside surface of the damping chamber, wherein a diameter of the damping section is greater than a diameter of the top section, wherein a diameter of the damping chamber is greater than a diameter of the actuating chamber so as to accommodate the damping section, and wherein the drive piston is included entirely within the housing such that, during operation of the drive piston, the damping section remains in the damping chamber, and the top section is at least partially disposed in the actuating chamber.

18. The HVA system of claim 17, wherein the at least one engine valve includes a first engine valve and a second engine valve each attached to a bottom of a bridge, and wherein a top of the bridge is attached to a bottom surface of the adjustment pin so as to control operation of the first and second engine valves in response to the movement of the adjustment pin.

19. The HVA system of claim 17, wherein the damping section is disk-shaped, and wherein the at least one surface of the damping section is a conical surface.

20. The HVA system of claim 17, wherein the at least one surface of the damping chamber is a cylindrical surface.
